# E · Steel rail fabrication and drilling

All steel work happens on the bench, before either rail goes near a top. Under the desk you only drive screws and bolts; you never drill steel there. Two identical lengths of 2 × 2 × 3/16 in hot-rolled angle arrive; either can be rail A. All dimensions in mm.

> **!** **MEASURE BEFORE MARKING OR DRILLING** MEASURE ON HARDWARE
> 1 · Tape each rail from end to end and write the length on the tape label. Design length **2130**; bought as 84 in = **2133.6**. If a rail is under 2130, stop and check before drilling.
> 2 · Calipers on each angle: legs 50.8, thickness 4.76 (the docs expect about 4.8).
> 3 · Every station is measured from the **labelled end**: rail A's **wall end**, rail B's **corner end**. The **far end carries any extra** length. Never centre the hole pattern on the bar and never shift a hole to share out the extra.

**1** **Label the datum ends.** Wrap tape round one end of each rail and write on it: **RAIL A · WALL END** on one, **RAIL B · CORNER END** on the other. Measure both rails from those ends (Fig. E-1). File both cut ends of each rail smooth.

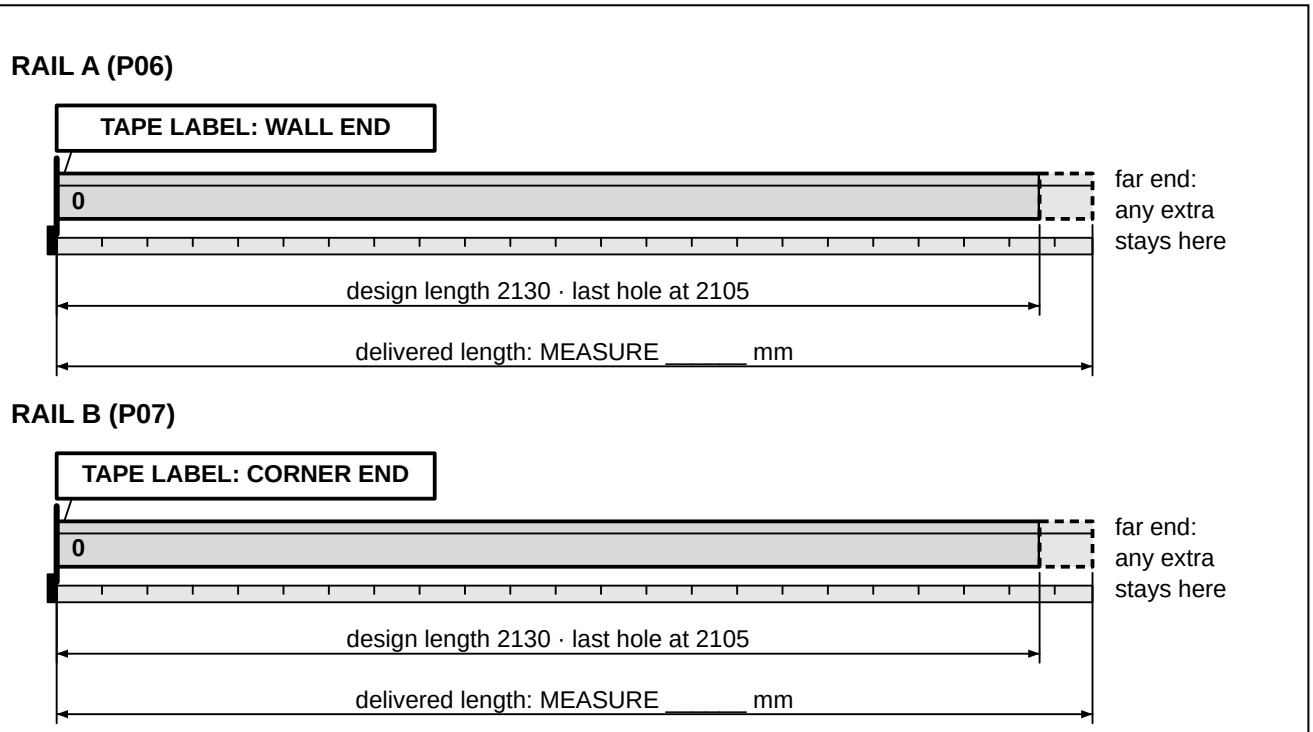



Fig. E-1 · The labelled end is 0 for every station. The last hole is at 2105 on both rails, so a longer rail only lengthens the far end beyond it. SOURCE-DEFINED design length; MEASURE ON HARDWARE delivered length.

| You need | For |
|---|---|
| P06 rail A, P07 rail B; one P08 fitting as the template | the work pieces |
| P18 printed tools: edge T-square, rail marking gauge, rail drill jig (PETG), 2 fitting slot plugs | stations, the 19 and 29 lines, pilots, bolt-hole transfer |
| Bits 1/8, 7/32 and 7/16 in (the 7/16 cobalt), cutting oil, flat file, 2 clamps, tape, combination square, pencil | drilling and deburring; a drill press if you have one |
| Acetone or mineral spirits, 120 grit, flat black enamel | finish (step 9) |

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## Which way up: the section and the two lines

**FLAT LEG**
lies against the top's underside

**VERTICAL LEG**
hangs down, toward you; outer face at
y = 530 (rail A)
x = 530 (rail B)

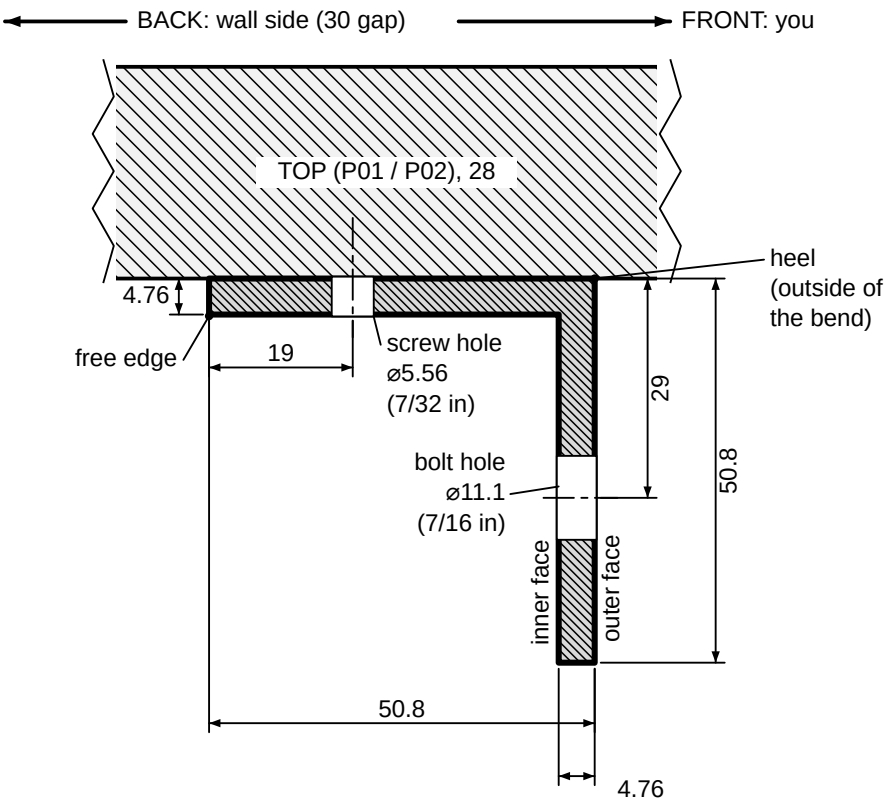
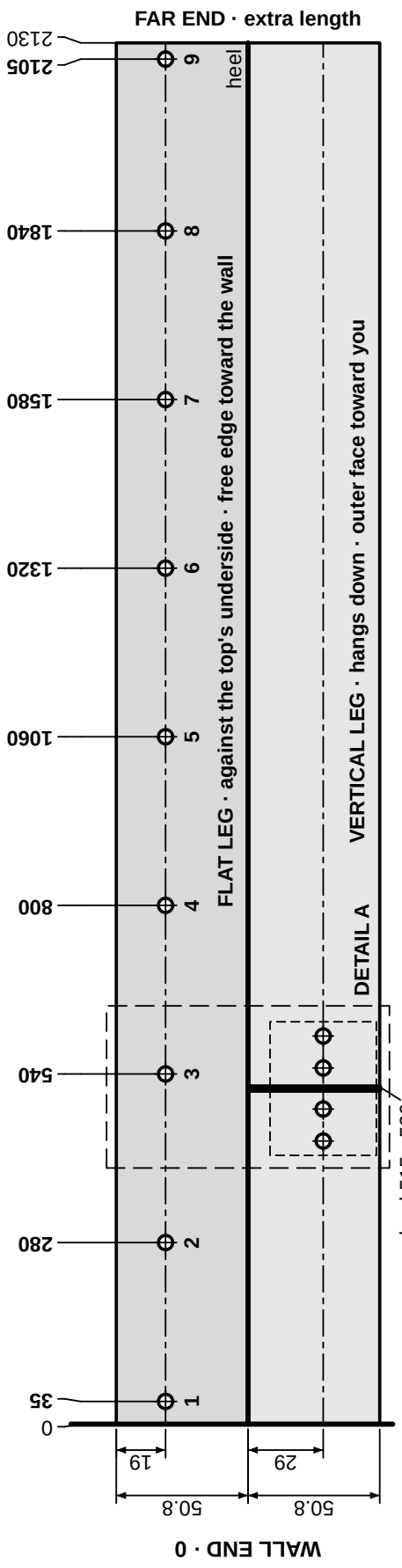



Fig. E-2 · Rail A seen end-on from its far end, as it hangs under top A (rail B is the same section turned 90°). Datum: top A's back-left corner, x along the back wall, y along the side wall. The vertical leg's outer face is at y = 530 on rail A and x = 530 on rail B, 120 back from the front edge (650).

| Item | mm | What it means | Confidence |
|---|---|---|---|
| Section | 50.8 × 50.8 × 4.76 | 2 × 2 × 3/16 in hot-rolled angle | SOURCE-DEFINED MEASURE ON HARDWARE thickness |
| Flat leg | | lies against the top's underside; its free edge points to the wall | SOURCE-DEFINED |
| Vertical leg | | hangs down on the side nearer you; outer face y = 530 (A), x = 530 (B) | SOURCE-DEFINED |
| Screw line | 19 | from the flat leg's free edge; centres of the ø5.56 (7/32 in) screw holes | SOURCE-DEFINED |
| Bolt line | 29 | below the heel (the top's underside); centres of the ø11.1 (7/16 in) bolt holes | SOURCE-DEFINED |
| Fitting slots | 21 × 12 | two per P08 leg, centres 29.15 and 78.45 from the heel; legs 100.5 × 41, 5 thick | VERIFIED 23 Oct 2026 |

**Bench set-up.** Clamp the rail with the flat leg's face up and the vertical leg hanging down over the bench edge. That is how it is marked and how the flat-leg holes are drilled. For the vertical leg (steps 6 and 7) turn it so the vertical leg's outer face is up and the flat leg hangs over the edge.

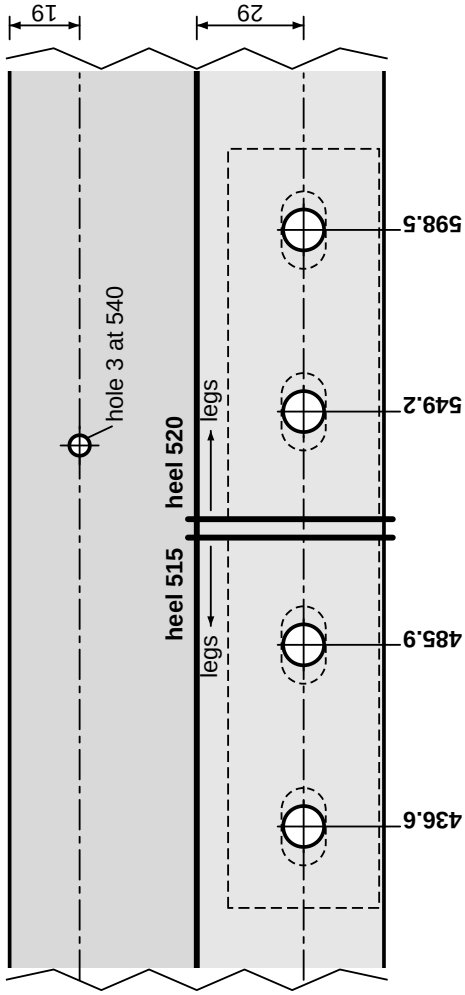
The two fabrication drawings that follow are full pages, turned sideways: turn the booklet a quarter turn clockwise to read them. Every hole is given as an ordinate from the labelled end, never from the hole before.




9× ø5.56 (7/32 in) THRU in the flat leg, centres on the screw line 19 from the free edge, at the ordinates above.
4× ø11.1 (7/16 in) THRU in the vertical leg on the bolt line 29 below the heel, at 436.6, 485.9, 549.2, 598.5 (Detail A).
**Positions VERIFIED from the measured fitting; drill them THROUGH the clamped fitting with the slot plug (step 6).**

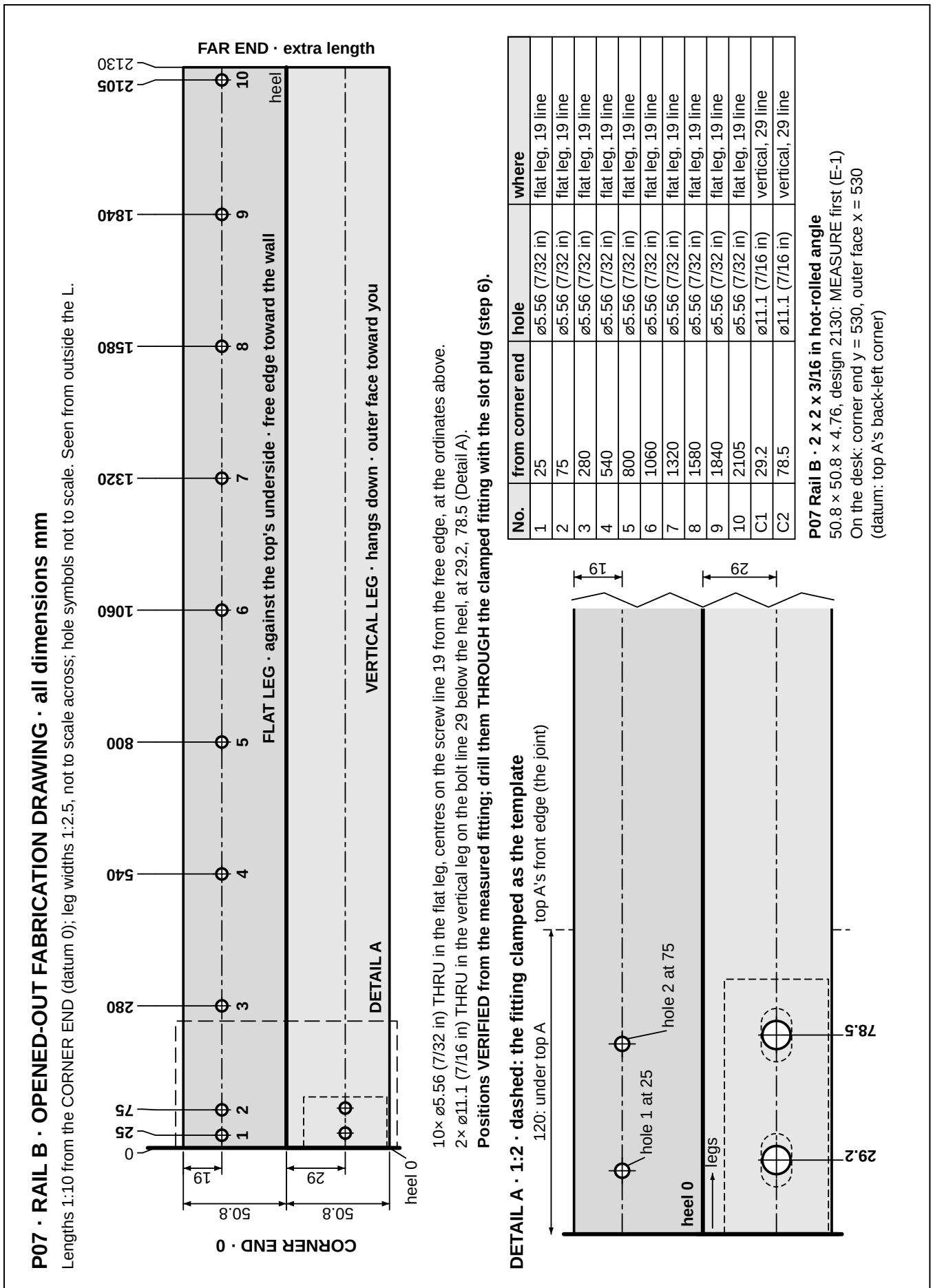
| No. | from wall end | hole | where |
|---|---|---|---|
| 1 | 35 | ø5.56 (7/32 in) | flat leg, 19 line |
| 2 | 280 | ø5.56 (7/32 in) | flat leg, 19 line |
| 3 | 540 | ø5.56 (7/32 in) | flat leg, 19 line |
| 4 | 800 | ø5.56 (7/32 in) | flat leg, 19 line |
| 5 | 1060 | ø5.56 (7/32 in) | flat leg, 19 line |
| 6 | 1320 | ø5.56 (7/32 in) | flat leg, 19 line |
| 7 | 1580 | ø5.56 (7/32 in) | flat leg, 19 line |
| 8 | 1840 | ø5.56 (7/32 in) | flat leg, 19 line |
| 9 | 2105 | ø5.56 (7/32 in) | flat leg, 19 line |
| I2 | 436.6 | ø11.1 (7/16 in) | vertical, 29 line |
| I1 | 485.9 | ø11.1 (7/16 in) | vertical, 29 line |
| O1 | 549.2 | ø11.1 (7/16 in) | vertical, 29 line |
| O2 | 598.5 | ø11.1 (7/16 in) | vertical, 29 line |

**DETAIL A · 1:2 · dashed: the fitting clamped as the template**




**P06 Rail A · 2 x 2 x 3/16 in hot-rolled angle**
50.8 × 50.8 × 4.76, design 2130: MEASURE first (E-1)
On the desk: wall end x = 10, outer face y = 530
(datum: top A's back-left corner)

Fig. E-3 · P06 rail A, opened out: the flat leg (upper strip) and the vertical leg (lower strip) unfolded about the heel. Nine screw holes from the wall end; four bolt holes at 436.6, 485.9, 549.2 and 598.5, drilled through the fitting (heels at 515 and 520). SOURCE-DEFINED stations; VERIFIED bolt holes.



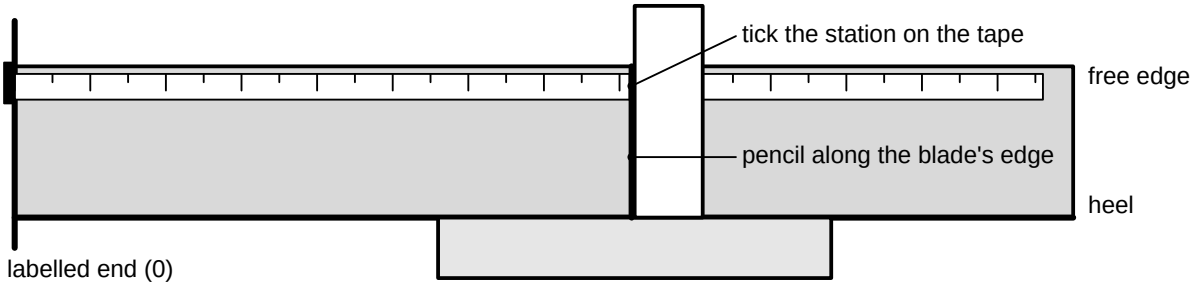

| No. | from corner end | hole | where |
|---|---|---|---|
| 1 | 25 | ø5.56 (7/32 in) | flat leg, 19 line |
| 2 | 75 | ø5.56 (7/32 in) | flat leg, 19 line |
| 3 | 280 | ø5.56 (7/32 in) | flat leg, 19 line |
| 4 | 540 | ø5.56 (7/32 in) | flat leg, 19 line |
| 5 | 800 | ø5.56 (7/32 in) | flat leg, 19 line |
| 6 | 1060 | ø5.56 (7/32 in) | flat leg, 19 line |
| 7 | 1320 | ø5.56 (7/32 in) | flat leg, 19 line |
| 8 | 1580 | ø5.56 (7/32 in) | flat leg, 19 line |
| 9 | 1840 | ø5.56 (7/32 in) | flat leg, 19 line |
| 10 | 2105 | ø5.56 (7/32 in) | flat leg, 19 line |
| C1 | 29.2 | ø11.1 (7/16 in) | vertical, 29 line |
| C2 | 78.5 | ø11.1 (7/16 in) | vertical, 29 line |

**P07 Rail B · 2 x 2 x 3/16 in hot-rolled angle**
50.8 × 50.8 × 4.76, design 2130: MEASURE first (E-1)
On the desk: corner end y = 530, outer face x = 530
(datum: top A's back-left corner)

Fig. E-4 · P07 rail B, opened out. Ten screw holes from the corner end: holes 1 and 2 go into top A (the first 120 sits under top A), holes 3 to 10 into top B. Two bolt holes at 29.2 and 78.5, shared by both fittings, drilled through the fitting with its heel flush with the corner end. SOURCE-DEFINED stations; VERIFIED bolt holes.

**2** **Mark the stations.** Hook the tape over the labelled end and tick every station on the flat leg from that one hook: rail A 35, 280, 540, 800, 1060, 1320, 1580, 1840, 2105; rail B 25, 75, 280, 540, 800, 1060, 1320, 1580, 1840, 2105. Square each tick across the leg along the T-square's blade, its lip hooked over the heel edge.

Hook the tape over the labelled end once and tick every station from that one hook; never measure from one hole to the next.



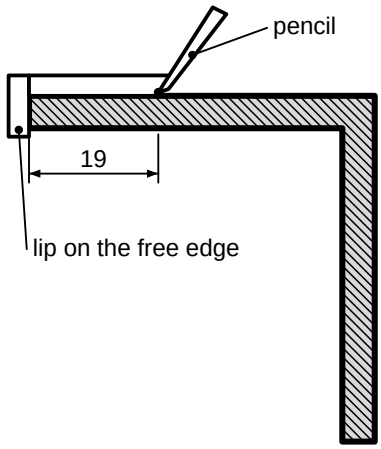

T-square (P18): lip over the heel edge, blade across the leg

Plan: flat leg face up, vertical leg over the bench edge, labelled end at the left.

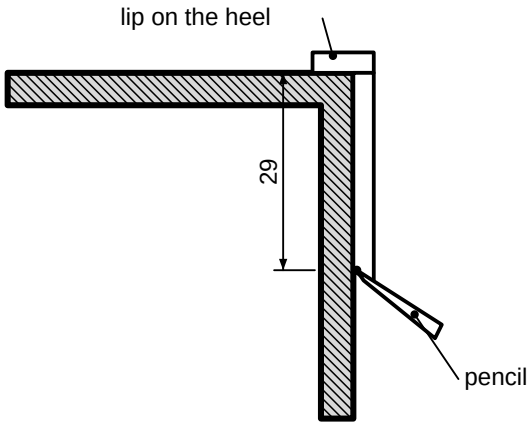
Fig. E-5 · One hook, every station. The station line only has to cross the leg; the jig finds the 19.

**3** **Pencil the two lines with the rail marking gauge, full length.** Screw line: lip over the flat leg's free edge, pencil in the **19** notch, slide. Bolt line: gauge on the vertical leg's outer face, lip over the heel, pencil in the **29** notch, slide. The bolt line is only a check: the fitting's holes must sit on it in step 6.

**a SCREW LINE on the flat leg**
lip over the free edge,
pencil in the 19 notch, slide




**b BOLT LINE on the vertical leg**
gauge on the outer face, lip over the heel,
pencil in the 29 notch, slide




Sections, marking gauge (P18) in use. Run each line the full length of both rails.

Fig. E-6 · The gauge's lip is the zero; the V notches put the pencil point on 19.0 and 29.0. SOURCE-DEFINED

**4** **Pilot the screw holes through the drill jig.** Jig lip down over the flat leg's free edge, sight grooves on the station line, hold it down and drill 1/8 in through its bore: square, on the 19 line, no punch needed. All 19 stations (9 on rail A, 10 on rail B). Use a drill press if you have one.

**5** **Open and deburr.** Jig off. Open every pilot to **⌀5.56 (7/32 in)** at low speed with a drop of oil on each; the bit follows the pilot. Deburr both faces with the file or a larger bit turned by hand.

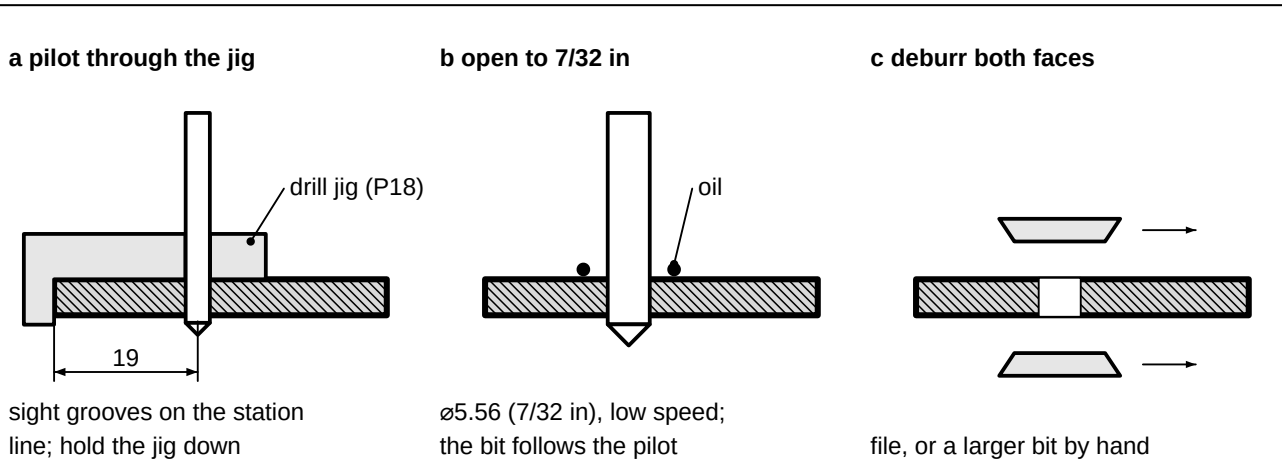



Fig. E-7 · Flat leg in section. The jig's bore is 1/8 in plus a running clearance; it keeps the bit square and on the 19 line.

**THE FITTING IS THE TEMPLATE** VERIFIED
The fittings were measured on 23 Oct 2026: legs 100.5 × 41, two 21 × 12 slots per leg with centres 29.15 and 78.45 from the heel. That puts the six bolt holes at the positions below. Even so, do not lay them out with the tape: clamp the fitting on its heel mark and drill through it with the printed **slot plug** (steps 6 and 7), so each hole lands on its slot's centre. Check first that a plug drops into each slot without rattle. The heel of a fitting is the **outside of its bend**.

| Clamp the fitting | Heel on | Legs toward | Holes made (mm from the labelled end) |
|---|---|---|---|
| Rail A, outside fitting | 520 from the wall end | the far end | O1 549.2, O2 598.5 |
| Rail A, inside fitting | 515 from the wall end | the wall end | I1 485.9, I2 436.6 |
| Rail B, one fitting (both share the holes) | the corner end itself (0) | the far end | C1 29.2, C2 78.5 |

Heel stations SOURCE-DEFINED; hole positions from the measured slot centres VERIFIED. Both fittings are identical, so holes copied from one fit the other.

**6** **Clamp the fitting and pilot through the plugs.** Rail with the vertical leg's outer face up. Clamp one fitting to the vertical leg with its heel exactly on the mark and its holes centred on the 29 line, legs the way Fig. E-8 shows. Drop a slot plug into each slot and drill the 1/8 in pilot through each plug's bore into the rail. Repeat for each of the three placements.

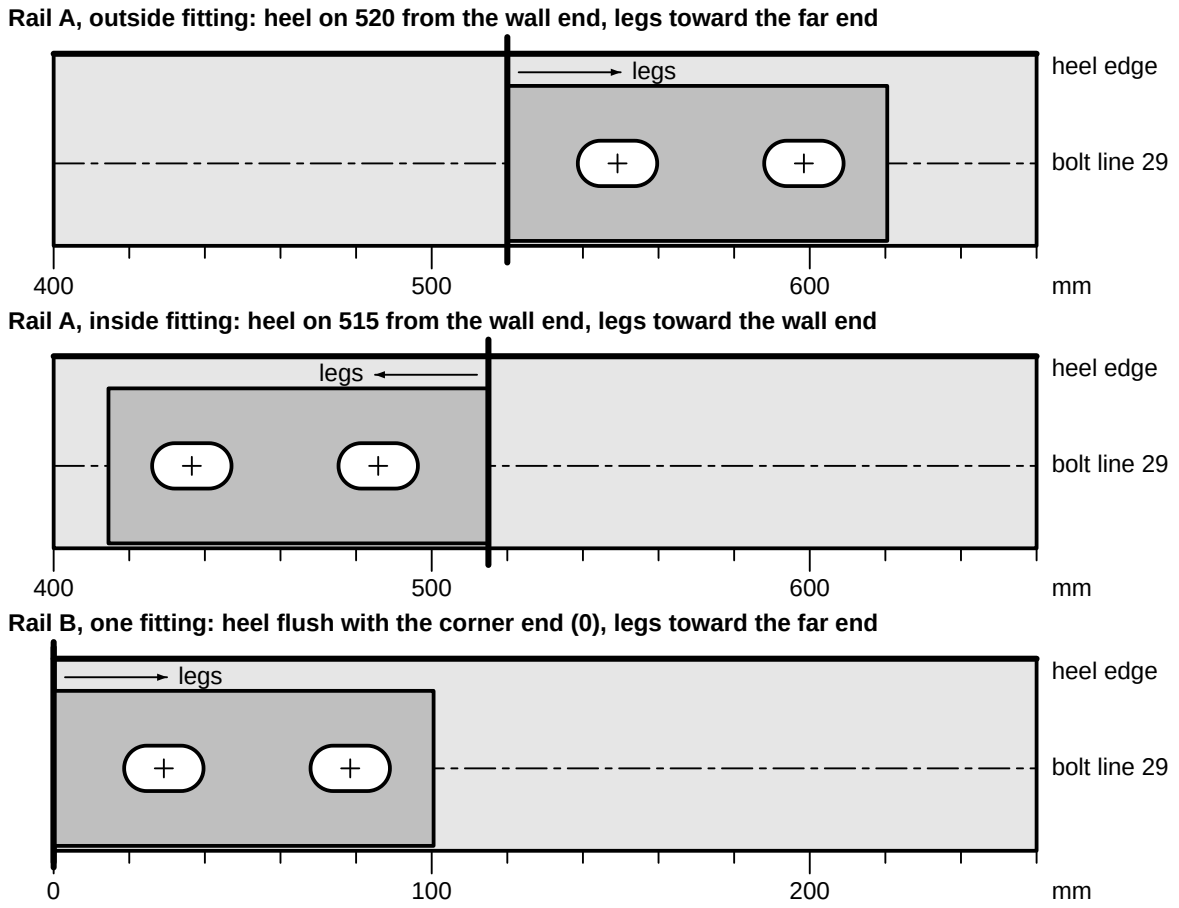



Fig. E-8 · The three clamp positions, vertical leg seen face on (1:2). Heavy line: the heel mark. Slots and their centres as measured. **VERIFIED**

**7** **Fitting off, open to ⌀11.1 (7/16 in).** Cobalt bit, slow, oil on each. The bit follows the pilot, so the hole lands on the slot's centre, not anywhere along its 21 mm. Deburr. Six holes: four in rail A, two in rail B.

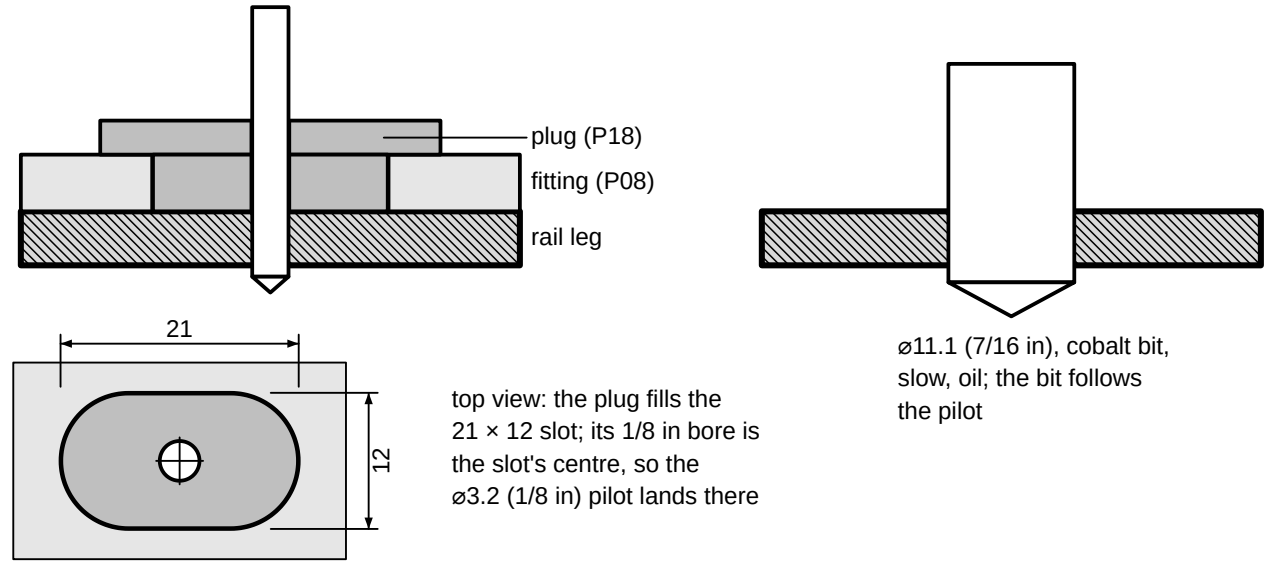
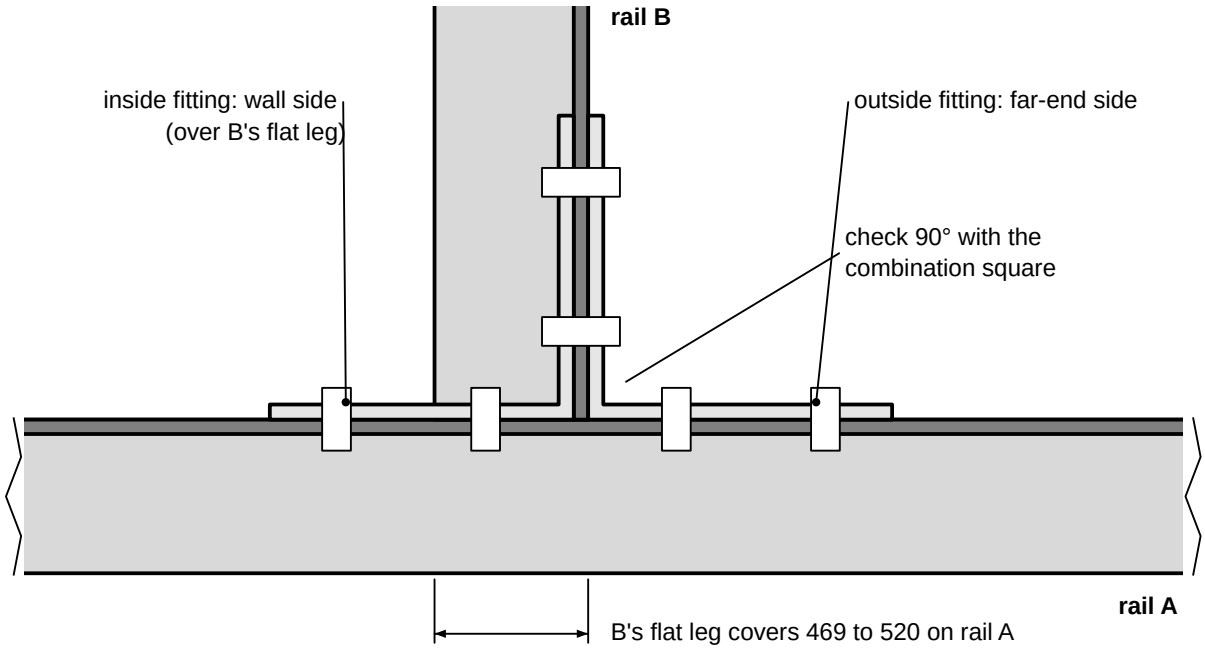



Fig. E-9 · The slot plug: a stadium that fills the 21 × 12 slot with a running clearance, 1/8 in bore at its centre. Without it a punch could land anywhere along the slot.

**8** **Floor test.** Lay both rails upside down on the floor, flat legs down, vertical legs up. Butt rail B's corner end against the outer face of rail A's vertical leg so that rail B's outer face is on rail A's **520** mark; rail B's flat leg then covers 469 to 520 on rail A's scale. Bolt both fittings on loosely: the outside one on the far-end side of rail B, the inside one on the wall side. Every bolt must slide in by hand. Check the corner with the combination square. If a bolt will not go in, open that hole with a round file now, not under the desk.

Plan from above. Both rails upside down on the floor: flat legs down, vertical legs up.
Bolts in loosely; every one must slide in by hand.




**B's outer face on rail A's 520 mark; B's end touching A: no gap**

Fig. E-10 · Floor test, plan from above. Bolts drawn as boxes. 469 = rail B's flat-leg edge (x = 479) less rail A's wall-end offset (x = 10). SOURCE-DEFINED

**9** **Finish.** Take the fittings off. Degrease both rails, scuff with 120 grit, two coats of flat black enamel, 24 h to cure. The holes need no masking.

The rails are now ready for the tops: rail A goes on top A with its wall end at x = 10 (1 cm in from the taped end), rail B on top B with its corner end projecting 120 past the joint end. Chapter F bolts the corner once both tops stand.

# F · Steel corner assembly

The two rails are joined at the inside corner by two gusseted strut fittings, one each side of rail B's end, bolted to the vertical legs. This bolted joint, not the joining plates, carries the inside corner of top B. Both tops are standing (build manual step 7); you work from below.

> **SEQUENCE: THE FITTINGS GO ON LAST** SOURCE-DEFINED
> Before the first fitting goes on, all of these must be done:
> 1 · The two surfaces are flush at the joint, leveled with the feet (never packed).
> 2 · Rail B's end touches the outer face of rail A's vertical leg with no gap, and rail B's outer face is on the 520 mark on rail A.
> 3 · Joining plates P1 and P2 are screwed across the joint. They align the tops; they do not carry the corner.
> 4 · Rail B's screws 1 and 2 are driven into top A. The inside fitting sits right under them and would block the driver.

| | Part | Qty | | Hardware | Qty |
|---|---|---|---|---|---|
| **P06** | Rail A, angle 2130, drilled (E) | 1 | **H08** | 3/8-16 × 1½ in hex bolt, grade 5 | 6 |
| **P07** | Rail B, angle 2130, drilled (E) | 1 | **H09** | 3/8 flat washer, under each head and nut | 12 |
| **P08** | Gusseted 90° strut fitting, 5 mm, legs 100.5 × 41, two 21 × 12 slots per leg VERIFIED | 2 | **H10** | 3/8-16 nylon-insert lock nut | 6 |

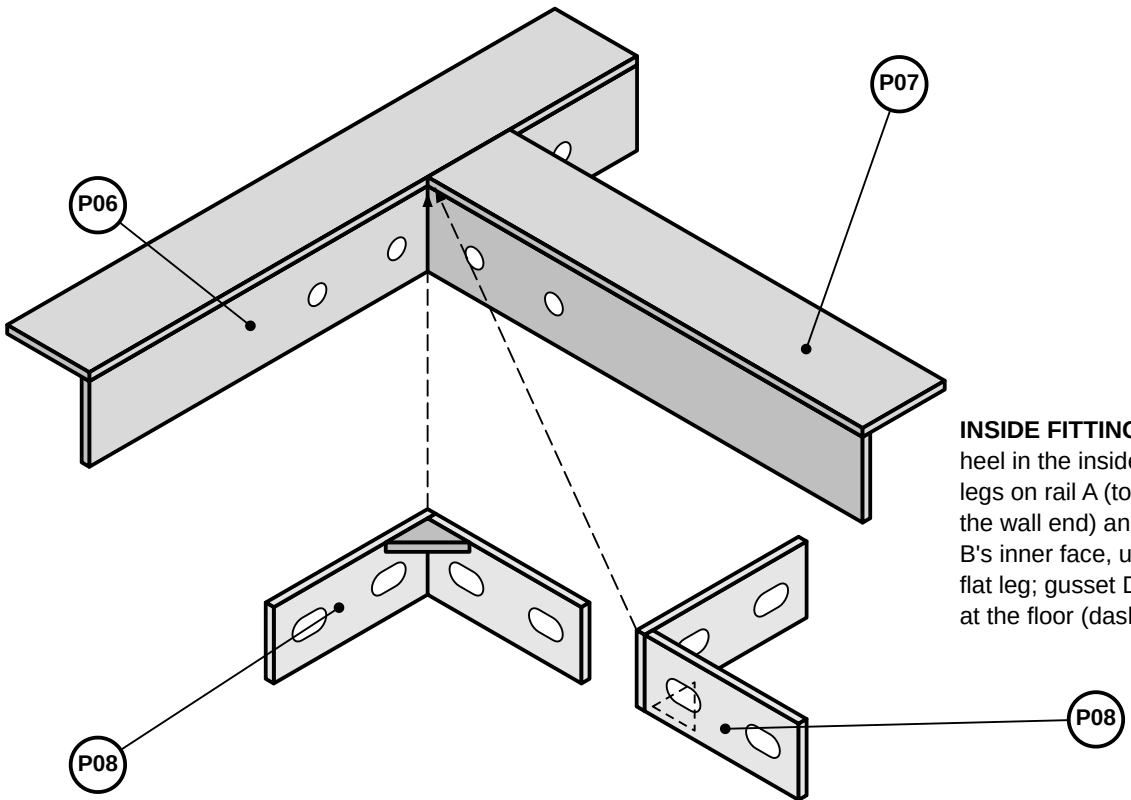



Fig. F-1 · The corner seen from the knee space with the tops lifted away. Both fittings are drawn pulled out of place; the dashed lines run from each heel to where it goes. The outside fitting fills the outside corner, the inside one the pocket under rail B's flat leg, gusset down. Neither is a mirror image: they are the same part, the inside one turned gusset-down. SOURCE-DEFINED placement.

## Which bolt goes where

| Bolts | Through | Head and washer | Washer and nut |
|---|---|---|---|
| **O1, O2** | outside fitting + rail A's vertical leg | fitting side, in the knee space | rail A's inner face, under its flat leg |
| **I1, I2** | inside fitting + rail A's vertical leg | fitting side, in the pocket under rail B | rail A's inner face, under its flat leg |
| **C1, C2** | outside fitting + rail B's vertical leg + inside fitting | outside fitting, knee side | inside fitting, in the pocket |

Every bolt goes in **from the fitting side**: four through rail A's leg with one fitting each, two through rail B's leg with a fitting on each side.

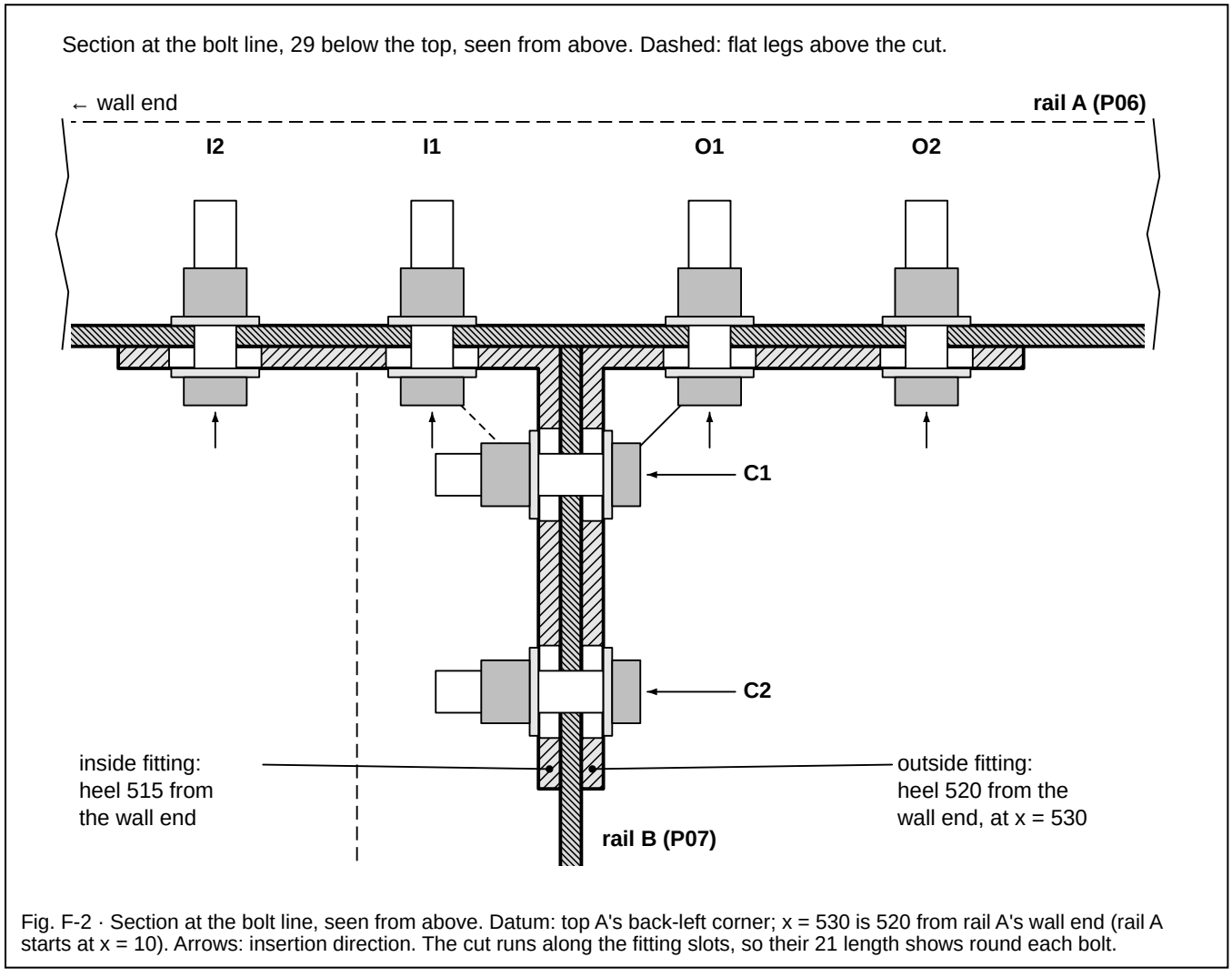


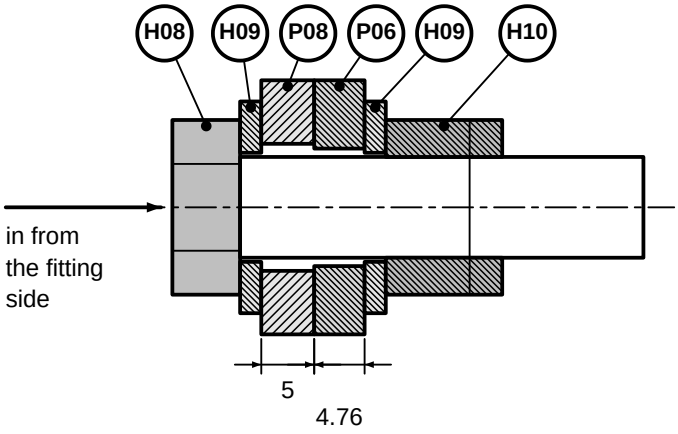

Fig. F-2 · Section at the bolt line, seen from above. Datum: top A's back-left corner; x = 530 is 520 from rail A's wall end (rail A starts at x = 10). Arrows: insertion direction. The cut runs along the fitting slots, so their 21 length shows round each bolt.

1 **Outside fitting.** Heel into the outside corner, one leg flat on rail A's vertical leg toward the far end, the other on rail B's outer face, gusset up, facing the knee space. Push O1 and O2 through from the fitting side with a washer under each head; washer and nut on rail A's inner face. Finger tight.

2 **Inside fitting.** Into the pocket: heel in the inside corner, one leg on rail A's vertical leg toward the wall end, the other on the inner face of rail B's vertical leg, **gusset pointing at the floor**. Gusset-up it would sit 1 mm under the head of rail B's screw 1. I1 and I2 from the fitting side, washers, nuts on rail A's inner face. Finger tight.

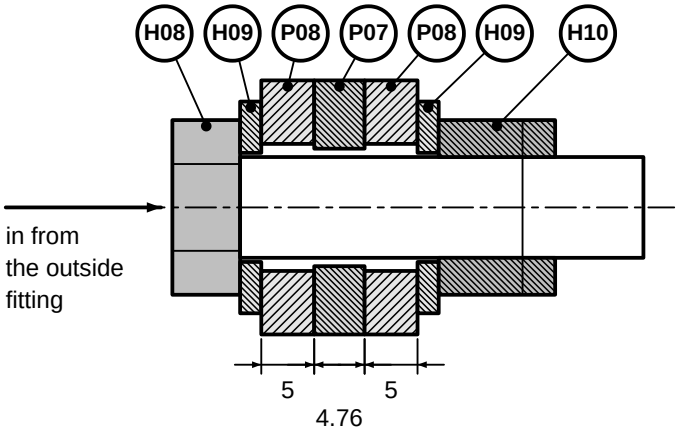
**3** **Through bolts.** C1 and C2 from the outside fitting: through the outside fitting, rail B's vertical leg and the inside fitting, washer under the head, washer and nut in the pocket. Finger tight. All six bolts are now in; none is tight.

**Rail A: 4 bolts, one fitting each (O1, O2, I1, I2)**




fitting 21 × 12 slot (cut across):
2.5 mm play across, 11.5 mm along
rail hole ⌀11.1 (7/16 in):
1.6 mm of play
**so the joint holds by friction:**
**tighten fully, 25 N·m**

**Rail B: 2 bolts through outside fitting, rail B and inside fitting (C1, C2)**




head and washer on the
outside fitting (knee side);
washer and nut on the
inside fitting, in the pocket
under rail B's flat leg

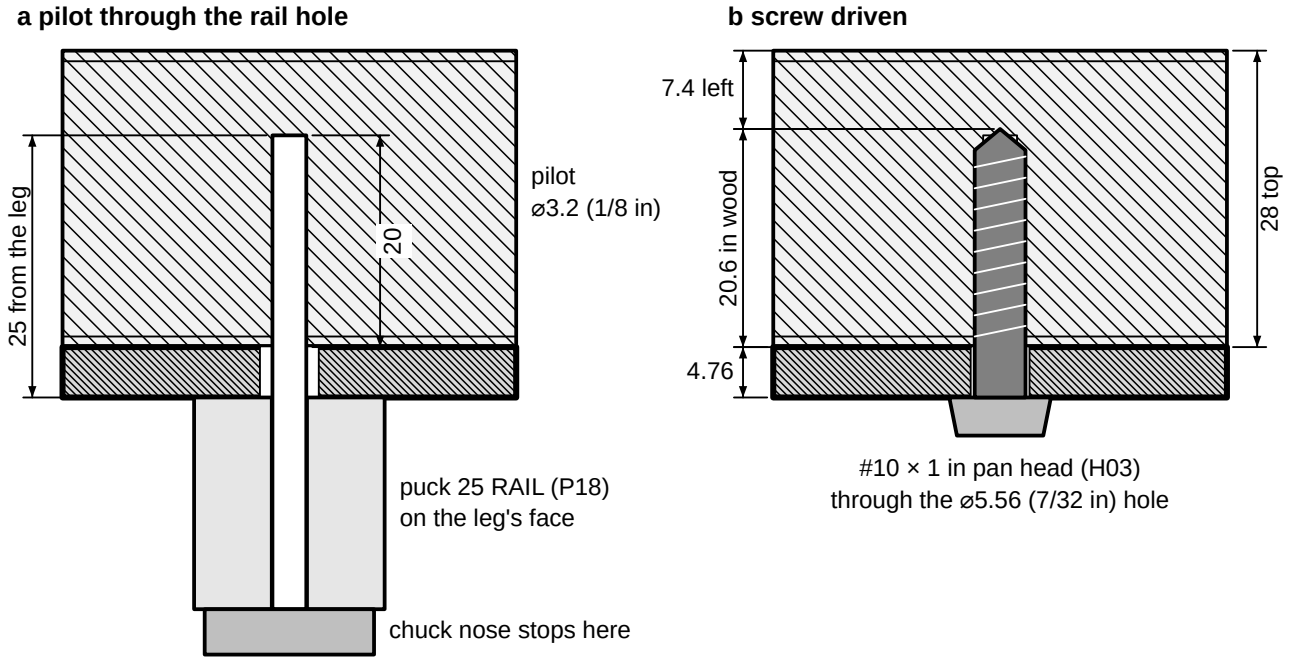
Fig. F-3 · Bolt stack-ups, cut along the bolt. Plates drawn cut off. The fitting's slots are 21 × 12; the cut shows their 12 width.
VERIFIED fitting; SOURCE-DEFINED bolts.

**4** **Check again, then tighten all six.** Fingertip across the joint: no step. Rail B's end still touching rail A. Then tighten every bolt fully:

! **TIGHTENING: 25 N·m (18 ft·lb)** SOURCE-DEFINED
With a torque wrench, **25 N·m (18 ft·lb)**, under the standard dry torque even for a grade 2 bolt. Without one, pull as hard as one hand can on a 15 cm wrench, then stop.
**Why tight, not snug:** this is a friction joint. The fittings' 21 × 12 slots leave 2.5 mm of play across and 11.5 mm along each bolt, the 7/16 in rail holes 1.6 mm, so only the clamping force keeps rail B's corner from settling.
**When:** only after the two surfaces are flush and rail B's end touches rail A.
**Later:** if the corner ever drops a millimetre, retighten these six bolts.

## How the rails hold the tops

Each rail is screwed to its top through the flat leg: nine #10 × 1 in pan-head Robertson screws (H03) in rail A, ten in rail B, 19 in all. Rail B's holes 1 and 2 land in top A (its first 120 lies under top A) and are driven in build-manual step 7.4, before the fittings; the rest go in with the tops face down.




Never more than 21 into the 28 top. Drawn upright, as for rail B's holes 1 and 2 (overhead).

Fig. F-4 · One rail screw in section. The pilot is 1/8 in, 25 from the flat leg's face (the face the puck stands on), which is 20 into the wood; the screw passes the 4.76 steel and goes 20.6 into the 28 top, leaving 7.4 under the laminate. SOURCE-DEFINED

| Item | mm | Source |
|---|---|---|
| Top thickness | 28 | SOURCE-DEFINED MEASURE ON HARDWARE calipers (if 27, shorten every screw into the top by 1) |
| Flat leg | 4.76 | SOURCE-DEFINED |
| Screw into the top | 20.6 | SOURCE-DEFINED; limit 21 into any top |
| Pilot depth from the leg's face | 25 | printed drill guide, puck 25 RAIL SOURCE-DEFINED |

**Settling check.** Lay a straightedge across the joint at the inside corner and write down any step. Check again after two weeks and after the first month; if the corner has settled, retighten the six fitting bolts.